Course Name:	Web Development Laboratory (316U01L306)	**Semester:**	**III**
Date of Performance:	**__12_ / __08_ / __26____**	**DIV/ Batch No:**	**B-2**
Student Name:	**Ishaan Singh Khanka**	**Roll No:**	**16010125107**

Experiment No: 2

Title: Design web Portal using CSS3.

Aim of the Experiment: To apply CSS3 style sheets to an existing HTML5-based Web Portal for the given Use Case study in order to enhance layout, responsiveness, visual aesthetics, and user interaction using modern CSS3 features.

Objectives for the Experiment:

1. Implement different CSS3 styling techniques
2. Design responsive layouts using Flexbox and media queries
3. Enhance user interaction using animations
4. Maintain separation of content and presentation

COs to be achieved:

CO1: Use CSS to prepare the layout of web pages.

Books/ Journals/ Websites references:

1. Students should write ……..

Theory:

Students should write about CSS3 (citation needed for content taken (IEEE or APA7 style))

Problem statement: Demonstrate the implementation of CSS3 features to enhance the layout, responsiveness, and aesthetics of the selected HTML-based web portal for the selected Use Case in experiment 1.

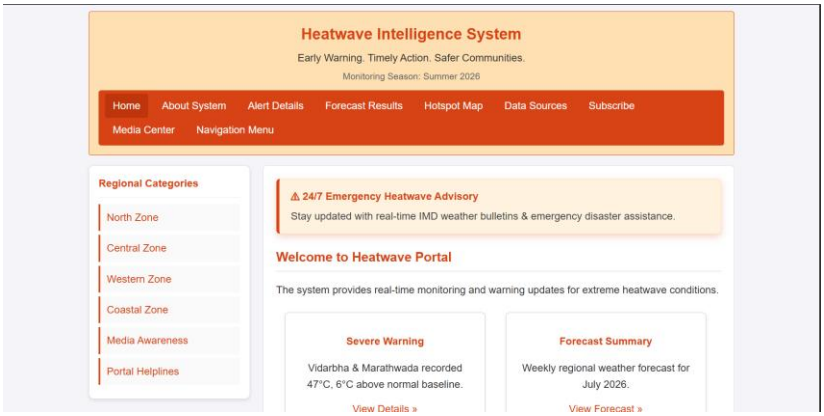
Note: All tasks are compulsory to use.

Task 1: Style the Homepage Header (Inline CSS)

Requirements:

- Use **inline CSS** to:
  - Set background color for header
  - Change text color of store name
  - Center align slogan
  - Add padding and border
- Apply inline style to:
  - <h1>, <p>, <header>

Concepts tested:
Inline CSS, color, background, padding, text-align, border.



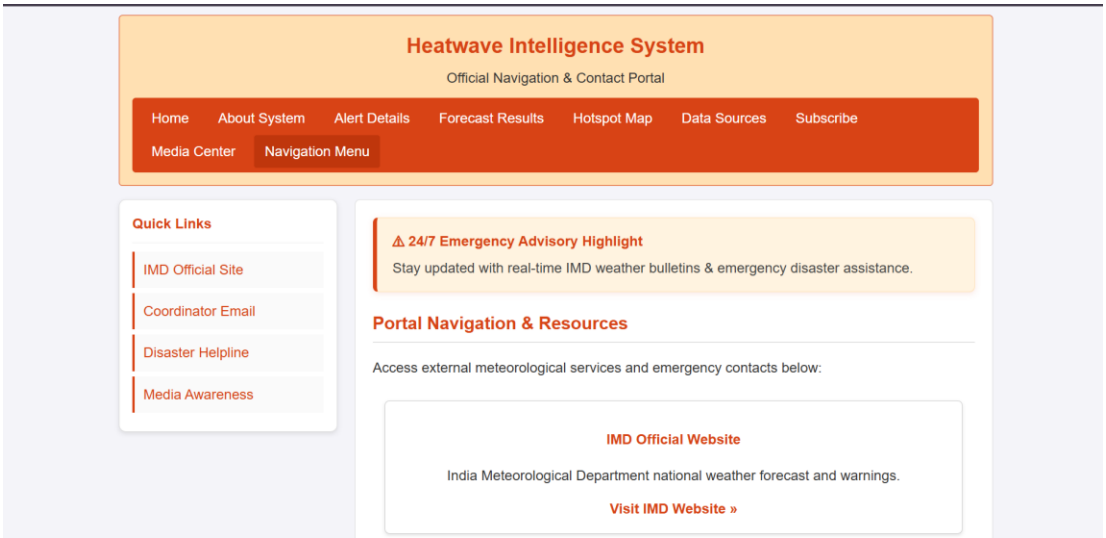
Task 2: Internal CSS for Navigation Menu

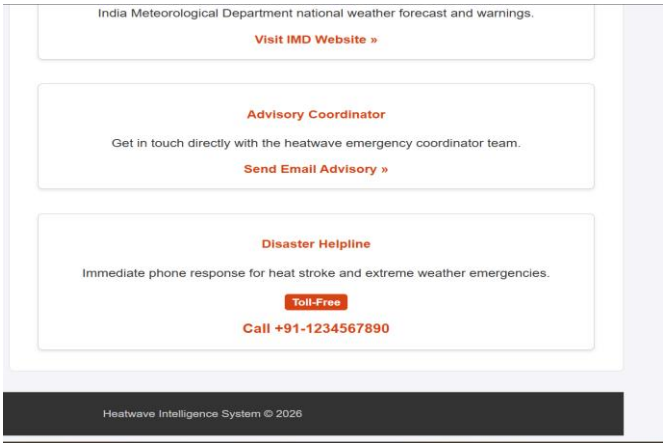
Navigation menu should look professional and consistent across the page.

Requirements:

- Use **internal CSS (<style> in <head>)**
- Style navigation menu:
  - Remove bullet points
  - Display menu items horizontally
  - Add hover effect
  - Change link colors
- Use:
  - :hover, text-decoration, display: inline-block

Concepts tested:
Internal CSS, selectors, pseudo-class, styling links.





Task 3: External CSS for Whole Website Theme

For a consistent look across all pages.

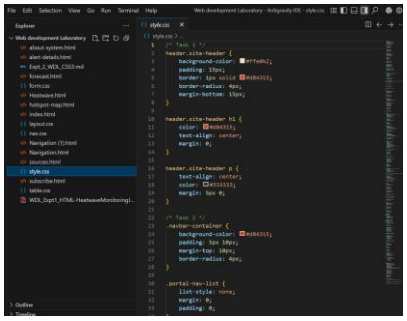
Requirements:

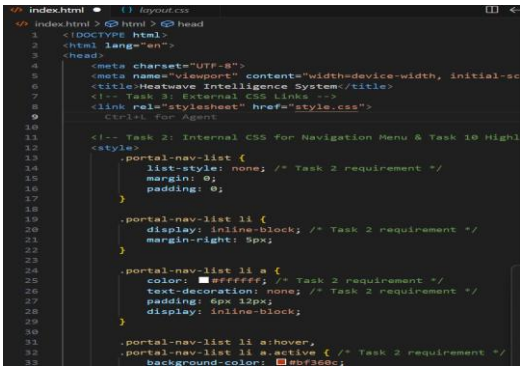
- Create file: style.css
- Link it using <link>
- Apply:
  - Body font-family
  - Background color
  - Common heading color

- Common footer style

Concepts tested:
External CSS, reusable styles, site-wide theming.

Deliverables : style.css linked with book.html





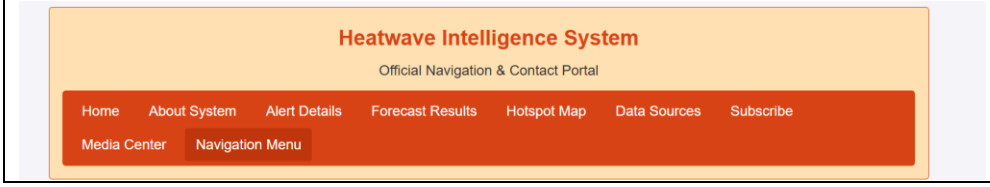
Task 4: Style Book Categories Sidebar

The categories section should look like a card.

Requirements:

- Use **external or internal CSS**
- Apply:
  - Border
  - Box shadow
  - Padding
  - Rounded corners
- Style list items with:
  - Spacing
  - Hover color change

Concepts tested:
Box model, border-radius, box-shadow, list styling.



Task 5: Book Card Layout Using CSS Box Model

To make Each item appear as a product card.

Requirements:

- Style a as a book card:
  - Width, margin, padding, border
  - Center align content
- Use:
  - display: inline-block
  - Add background color and shadow

Concepts tested:
Box model, margin, padding, display property.

Deliverable : link this .css file with book.html

```css
/* Task 5 */
.feature-card, .book-card {
    display: inline-block;
    width: 45%;
    margin: 10px 2%;
    padding: 15px;
    border: 1px solid #ccc;
    border-radius: 6px;
    background-color: #ffffff;
    box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
    text-align: center;
    vertical-align: top;
    box-sizing: border-box;
}

.feature-card:hover, .book-card:hover {
    border-color: #d84315;
}

.nav-card-container {
    display: flex;
    flex-wrap: wrap;
    gap: 15px;
    margin-top: 15px;
}

.nav-card {
    flex: 1 1 calc(50% - 15px);
    min-width: 250px;
}

.helpline-badge {
```

Task 6: Image Gallery Styling with CSS

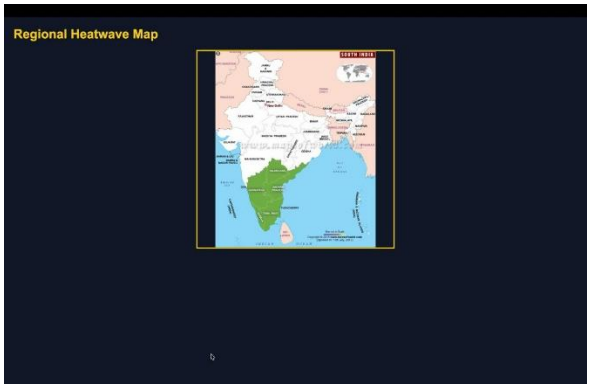
To make Featured appropriate items/ attributes, whichever applicable in the Use Case look like a grid.

Requirements:

- Use **CSS** to:
  - Set same size for all images
  - Add border and hover zoom effect
  - Add spacing between images
- Use:
  - transition, transform: scale()

Concepts tested:
Image styling, hover effects, transitions.

Deliverable nav.css , link to nav bar of expt1



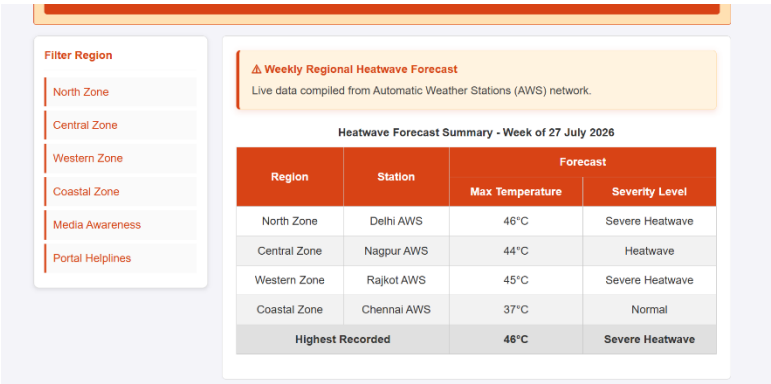
Task 7: Table Styling for Price List

Requirements:

- Use CSS to:
  - Collapse borders
  - Add header background color
  - Zebra-striping rows
  - Center align text
- Use:
  - border-collapse, nth-child(even)

Concepts tested:
Table styling, pseudo-classes.

Deliverable table.css , link to the corresponding file.




```css
/* Task 7 */
table {
    width: 100%;
    border-collapse: collapse;
    margin: 15px 0;
}

table caption {
    font-size: 16px;
    font-weight: bold;
    margin-bottom: 8px;
}

table th, table td {
    border: 1px solid #cccccc;
    padding: 8px 12px;
    text-align: center;
}

table th {
    background-color: #d84315;
    color: #ffffff;
}

table tbody tr:nth-child(even) {
    background-color: #f2f2f2;
}

table tfoot td {
    background-color: #e0e0e0;
    font-weight: bold;
}
```

Task 8: Style Registration Form

Requirements:

- Style:
  - Input boxes width, padding
  - Submit button color and hover effect
  - Fieldset and legend
- Use:
  - :focus, :hover

- border-radius

Concepts tested:
Form UI styling, pseudo-classes, layout.

Deliverable : form.css



```css
fieldset {
    border: 1px solid #cccccc;
    border-radius: 6px;
    padding: 15px;
    margin-bottom: 15px;
}

legend {
    font-weight: bold;
    color: #d84315;
    padding: 0 5px;
}

label {
    display: block;
    margin-top: 10px;
    margin-bottom: 4px;
}

input[type="text"],
input[type="email"],
input[type="tel"],
select {
    width: 100%;
    padding: 8px;
    border: 1px solid #ccc;
    border-radius: 4px;
    box-sizing: border-box;
}

input[type="text"]:focus,
input[type="email"]:focus,
```



```css
}

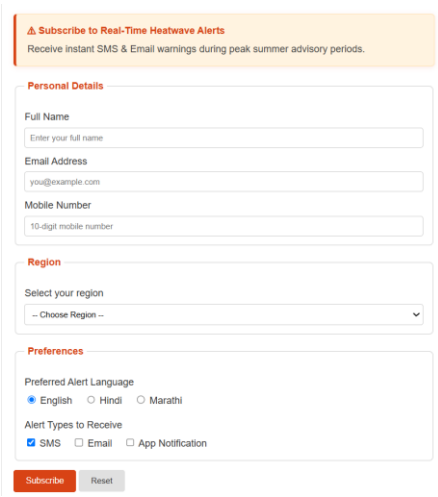
.inline-option {
    margin-right: 15px;
}

button[type="submit"] {
    background-color: #d84315;
    color: #ffffff;
    border: none;
    padding: 10px 20px;
    border-radius: 4px;
    cursor: pointer;
    font-size: 14px;
}

button[type="submit"]:hover {
    background-color: #bf360c;
}

button[type="reset"] {
    background-color: #e0e0e0;
    color: #333333;
    border: none;
    padding: 10px 20px;
    border-radius: 4px;
    cursor: pointer;
}

button[type="reset"]:hover {
    background-color: #cccccc;
}
```



Task 9: Page Layout Using CSS (Header, Sidebar, Content, Footer)

To Create a proper website layout.

Requirements:

- Use CSS to:
  - Set sidebar to left
  - Content to right
  - Footer at bottom
- Use:
  - float or flex
  - Set widths and spacing

Concepts tested:
Layout design, positioning, float/flex.

Deliverable : layout.css

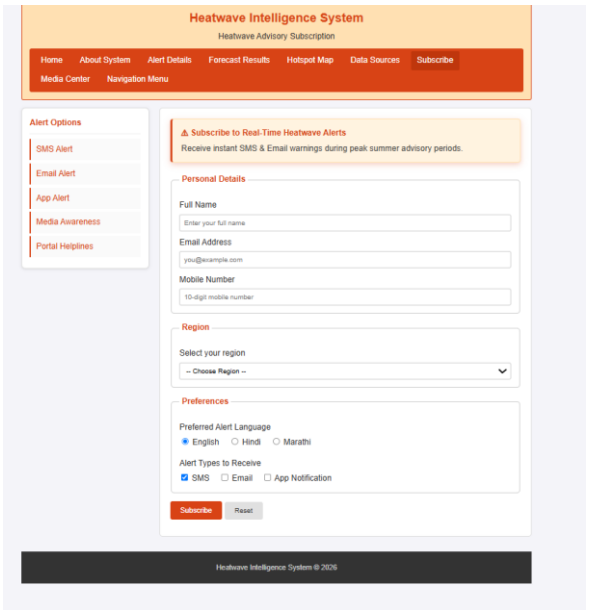
```css
/* Task 9 */
.page-container {
    max-width: 1000px;
    margin: 0 auto;
    padding: 10px;
}

.main-layout {
    display: flex;
    gap: 20px;
}

aside.sidebar {
    width: 250px;
    flex-shrink: 0;
}

main.main-content {
    flex-grow: 1;
    background-color: #ffffff;
    padding: 20px;
    border: 1px solid #ddd;
    border-radius: 6px;
}

@media (max-width: 700px) {
    .main-layout {
        flex-direction: column;
    }
    aside.sidebar {
        width: 100%;
    }
}
```



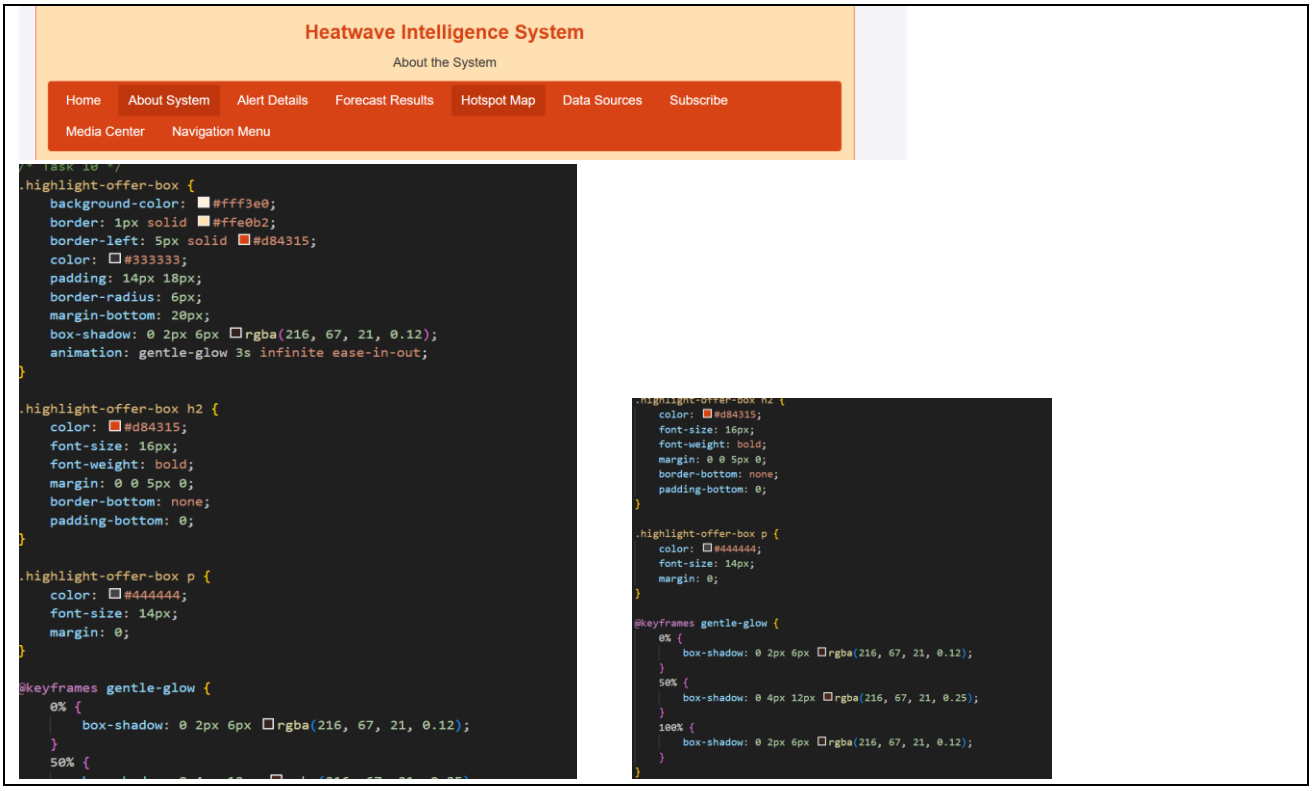
Task 10: CSS Effects & Highlighting.

To attract attention for Special events/data/section, etc..

Requirements:

- Create an “Offer” box:
  - Blinking or glowing effect
  - Different background color
  - Bigger font
- Use:
  - @keyframes animation OR hover effect
  - text-shadow, box-shadow

Concepts tested:
CSS animations, effects, visual emphasis.



Code:

Output / Screenshot:

Post Lab Subjective/Objective type Questions:

1. Use Bootstrap for CSS.

Conclusion and Discussion:

(students should write in your words)
(Also get feedback from fellow classmates (minimum 3) and faculty to add here)